AP® STATISTICS
2017 SCORING GUIDELINES

Question 4

Intent of Question

The primary goals of this question were to assess a student's ability to use boxplots to (1) compare multiple sets of data; (2) identify which set of data is most likely to have produced a particular summary value; and (3) determine which variable is most useful for classifying a new observation.

Solution

Part (a):

The median value for the percent of chemical Z in the pottery pieces is similar for all three sites, at about 7 percent. The ranges for the percent of chemical Z are much different for the three sites, with the smallest range of about 2 percent (from 6 percent to 8 percent) at Site II, a range of about 6 percent (from about 4 percent to 10 percent) at Site I, and the largest range of about 8 percent (from about 3 percent to 11 percent) at Site III.

Part (b):

(i) The piece most likely originated at Site III. Although values outside of the range of data observed in the samples would be possible, using the available data results in approximate minimum and maximum sums of the percents for the three chemicals as shown in the table below. Site III is the only site in which 20.5 falls between the sums of the minimum and maximum values.

	Site I		Site II		Site III	
Chemical	Min	Max	Min	Max	Min	Max
X	6	8	5	7	5	7.5
Y	11	15	1.9	4	6	8
Z	4	10	6	8	3	11
Sum	21	33	12.9	19	14	26.5

(ii) Chemical Y would be most useful, because the distribution of the percentages of total weights at the three sites do not overlap. The distributions of chemicals X and Z have substantial overlap.

Scoring

This question is scored in three sections. Section 1 consists of part (a), section 2 consists of part (b-i), and section 3 consists of part (b-ii). Each section is scored as essentially correct (E), partially correct (P), or incorrect (I).

Section 1 is scored as follows:

Essentially correct (E) if the response includes the following three components:
1. Recognition that the medians or centers are almost the same for the three sites
2. Recognition that the variability (ranges, IQRs, spread) is different across the three sites
3. Context is included

Partially correct (P) if the response includes only two of the three components.

Incorrect (I) if the response includes at most one of the three components.

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Question 4 (continued)

Notes:

- In all sections, comments about shape should be ignored because complete shape information is not obtainable from boxplots.
- Responses are not required to give numerical values. If responses provide numerical values, any reasonable approximation from the boxplots is acceptable.
- Because the boxplots are all symmetric, it is acceptable if the response discusses means instead of medians.
- Any discussion of chemical X and chemical Y is considered extraneous.
- Context is satisfied by any of the following references: site, chemical, weight, total weight, X, Y, or Z.

Section 2 is scored as follows:

Essentially correct (E) if the response includes the following three components:

1. Site III is chosen.
2. Sums of the minimum and maximum are computed for the three chemicals at each site.
3. A reasonable numerical justification is given involving sums of a statistical measure across the three chemicals to choose Site III.

Partially correct (P) if the response includes only two of the three components.

Incorrect (I) if the response includes at most one of the three components.

Notes:

- If the response computes only the sum of the minimums for Site I and the sum of the maximums for Site II and recognizes that this is sufficient, the response is scored E.
- If an alternative measure is used that involves sums of the three chemicals, such as the sum of the medians or the sums of the first quartiles and sums of the third quartiles, instead of the minimum and maximum sums, the second component is not satisfied, but the third component might be satisfied.
    - If the response explicitly or implicitly compares the alternate sum to the other two sites (for example, by indicating that the sum is the closest to 20.5 percent or by listing the sums for all three sites) the response is scored P.
    - If the response does not have an implicit or explicit comparison, the response is scored I.
- If either Site I or Site II is identified as the correct choice, no matter how that choice is justified, the response is scored I.
- The approximate sums of the medians are 27.5 for Site I, 16 for Site II, and 20 for Site III.

Section 3 is scored as follows:

Essentially correct (E) if the response chooses chemical Y *AND* gives a reasonable justification based on the fact that the distributions of chemical Y are distinctive across sites.

Partially correct (P) if the response chooses chemical Y *AND* provides justification based on the boxplots, but does not clearly explain that the distributions of chemical Y are distinctive across sites;
OR
if the response correctly discusses that the distributions of chemical Y are distinctive across sites, but never explicitly chooses chemical Y as the best choice, for instance, by stating only that there is substantial overlap across sites for chemicals X and Z but no overlap for chemical Y.

Question 4 (continued)

Incorrect (I) if the response does not meet the criteria for E or P.

Notes:

- To justify that the distributions of chemical Y are distinctive across sites, the justification must address both location and variability of the boxplots; for example, by stating that the boxplots do not overlap for chemical Y.
- If the response chooses chemical X or Z *OR* chooses chemical Y with no reasonable justification, the response is scored I.
- The justification that the distributions of chemical Y are distinctive across sites:
    - The following are acceptable because both location and variability are addressed. Such responses are scored E.
        - The boxplots for chemical Y do not overlap, or the boxplots for chemicals X and Z overlap.
        - All values of Site I are high, all values of Site II are low, and all values of Site III are in the middle.
        - The ranges never intersect.
        - The boxplots share no data.
        - Has completely different percentages at each site.
    - The following are incomplete justifications and are scored P.
        - The boxplots vary.
        - Chemical Y varies the most.
        - Chemical Y has the greatest variation.
        - The variation between/among sites is the largest.
        - The boxplots are different.
        - The medians/means differ.
        - The medians/means are most variable.
        - There is a difference in the percentages of chemical Y for each site.
        - The distribution of percents differs the most among the sites.

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Question 4

Intent of Question

The primary goals of this question were to assess a student's ability to (1) describe why the median might be preferred to the mean in a particular context; (2) compare the relative merits of two sampling plans; and (3) describe a consequence of nonresponse in a particular study.

Solution

Part (a):

The median is less affected by skewness and outliers than the mean. With a variable such as income, a small number of very large incomes could dramatically increase the mean but not the median. Therefore, the median would provide a better estimate of a typical income value.

Part (b):

Method 2 is better than Method 1. A sample obtained from Method 1 could be biased because of the voluntary nature of the response. It is plausible that class members with larger incomes might be more likely to return the form than class members with smaller incomes. The mean income for such a sample would overestimate the mean income of all class members. With Method 2, despite the smaller sample size, the random selection is likely to result in a sample that is more representative of the entire class and produce an unbiased estimate of mean yearly income of all class members.

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Question 4 (continued)

Scoring

This question is scored in three sections. Part (b) has two components: (1) identifying a relevant characteristic for each sampling method; (2) indicating the effect of the biased method on the estimate of the mean income. Section 1 consists of Part (a); section 2 consists of part (b), component 1; and section 3 consists of part (b), component 2. Sections 1, 2, and 3 are scored as essentially correct (E), partially correct (P), or incorrect (I).

Section 1 is scored as follows:

Essentially correct (E) if the response includes the following two components:

1. Describes how skewness or outliers affect the mean or do not affect the median.
2. Makes a conjecture about a relevant characteristic of the distribution of incomes, such as skewness or an outlier.

Partially correct (P) if the response includes only one of the two components listed above.

Incorrect (I) if the response does not meet the criteria for E or P.

Notes:

- For Component 1, examples of responses that are acceptable include:
  - The mean is affected by skewness (outliers).
  - The median is not affected by skewness (outliers).
  - The mean is greater (less) than the median when there is right (left) skewness or outliers.
- For Component 1, examples of responses that are not acceptable include:
  - Don't use the mean for skewed distributions or distributions with outliers.
  - Use the median for skewed distributions or distributions with outliers.
  - Responses that include an incorrect statement about means and/or medians, such as for right skewed distributions, the median will be higher than the mean.
- It is possible to satisfy *both* components with a single sentence, such as, "If there was a billionaire in the sample, the mean would be higher than the median."
- If a response argues that using the *mean* is a more appropriate way to estimate the typical income, then reduce the score in section 1 by one level (that is, from E to P or from P to I).

Section 2 is scored as follows:

Essentially correct (E) if the response chooses Method 2 *AND* includes the following two components:

1. Identifies a relevant characteristic of Method 1.
2. Identifies a relevant characteristic of Method 2.

Partially correct (P) if the response chooses Method 2 *AND* includes only one of the two components listed above

OR

if the response includes both components but does not choose a method.

Incorrect (I) if the response chooses Method 1 *OR* otherwise does not meet the criteria for E or P.

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Question 4 (continued)

Notes:

- Responses that do not explicitly choose Method 2 can still earn an E for section 2 if the choice is clearly implied. The choice of Method 2 is clearly implied if the response only discusses negative characteristics of Method 1 and only discusses positive characteristics of Method 2, such as, Method 1 is biased but Method 2 uses a random sample.
- Responses that *compare* the two methods can satisfy *both* components, such as, saying that Method 1 is more biased or that nonresponse will be less of a problem with Method 2.
- Responses that refer to the nonresponse bias as *voluntary response bias*, *response bias, undercoverage* can still earn an E.
- Discussions of conditions for inference should be considered extraneous and ignored.

Section 3 is scored as follows:

Essentially correct (E) if the response includes the following two components:

1. Indicates the incomes of responders may be different from the incomes of nonresponders.
2. Indicates the biased sampling method may produce a misleading estimate/conclusion about the mean income, including direction, for example, "The sample mean is likely to be higher than the mean of the population."

Partially correct (P) if the response provides only one of the two components listed above.

Incorrect (I) if the response does not meet the criteria for E or P.

Notes:

- A single sentence can satisfy the first component of section 2 and the first component of section 3. (For example, "In method 1, rich people are more likely to respond.")
- For component 2, either direction is acceptable but the direction must be consistent with the identified bias. Saying only that Method 2 will be more accurate or more representative does not satisfy component 2.
- If a response addresses possible nonresponse bias in Method 2, the response can still satisfy both components of section 3.
- Responses that focus on the larger sample size in Method 1 can satisfy component 2 if such responses describe the effect as reducing the variability of the estimate. (For example, "I would use Method 1 since the larger sample size would give less variability of the mean.")
- Responses that focus on untruthful survey answers can satisfy component 2 if the effect on the estimate is appropriate. (For example, "People contacted in Method 2 might say they make more money than they actually do. This would make the estimated mean too high.")

Question 2: Focus on Collecting Data — 4 points

General Scoring Notes

- Each part of the question (indicated by a letter) is initially scored by determining if it meets the criteria for essentially correct (E), partially correct (P), or incorrect (I). The response is then categorized based on the scores assigned to each letter part and awarded an integer score between 0 and 4 (see the table at the end of the question).
- The model solution represents an ideal response to each part of the question, and the scoring criteria identify the specific components of the model solution that are used to determine the score.

	Model Solution	Scoring
(a)	Treatments: New drug, placebo. Experimental units: The 72 people who receive the new drug or placebo. Response variable: Improvement in acne severity	**Essentially correct (E)** if the response satisfies the following three components: 1. Identifies the treatments as new drug and placebo 2. Identifies the experimental units as the 72 people (subjects, participants, twins) in the experiment 3. Identifies the response variable as the improvement in acne severity **Partially correct (P)** if the response satisfies only two of the three components. **Incorrect (I)** if the response does not satisfy the criteria for E or P.

Additional Notes:

- To satisfy component 1, identification of the treatments must include both the placebo and the new drug.
- To satisfy component 2, the response must indicate that the experimental units are individual people. The response could refer to participants, subjects, twins, or members of the pairs of twins without explicitly mentioning the number 72. However, a response that states or implies that there are 36 experimental units (e.g., "the pairs of twins") does not satisfy component 2.
- To satisfy component 3, the response must include the context of "acne" and "improvement" (e.g., "improvement in acne severity," "acne improvement score"), but it does not need to include a reference to the scale, the dermatologist, two-week time periods, or treatments. Reasonable synonyms for improvement can be used, such as using "reduction" or "change" or by including the verbal descriptions of the scale ("no improvement" to "complete cure"). However, a description of a binary outcome (e.g., "whether or not the acne improves") does not satisfy component 3.
- For responses that indicate the 36 pairs of twins are the experimental units, component 3 may be satisfied by indicating that the response variable is the improvement in acne severity or by indicating that the response variable is the difference in improvement in acne severity.
- If the response provides parallel solutions (i.e., two or more complete solutions without choosing or indicating which is to be scored), the response is scored based on the weaker of the two solutions. For example, if a response says that the experimental units are "the 72 participants and the scores from 0 to 100," component 2 is not satisfied.

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	Model Solution	Scoring
(b)	Improvement scores will vary due to many factors, including initial acne severity, what treatment is received, and other variables such as diet and genetics. Because the pairs of twins are similar in initial acne severity, pairing allows for the variation in improvement scores due to the treatment received to be distinguished from variation due to initial acne severity, unlike in a completely randomized design. Consequently, using the matched-pairs design will provide a more precise estimate of the mean difference in improvement in acne severity for the new drug compared to the placebo and make it easier to find convincing evidence that the new drug is better, if it really is better.	**Essentially correct (E)** if the response describes a statistical advantage of a matched-pairs design AND satisfies the following three components: 1. The advantage pertains to an inference made after collecting the data (e.g., the ability to distinguish between the effects of the treatments or the precision of the estimate of the drug effect) 2. Indicates that the matched-pairs design is better by using a comparative word (e.g., easier, clearer, greater) or by making an explicit comparison to a completely randomized design 3. Includes context (e.g., "drug," "improvement," "acne," or "twins") **Partially correct (P)** if the response describes a statistical advantage of a matched-pairs design AND satisfies one or two of the three components. **Incorrect (I)** if the response does not satisfy the criteria for E or P.

Additional Notes:

- To be considered an advantage of a matched-pairs design, the advantage described must be true for a matched-pairs design and not be true for a completely randomized design. For example, saying that "random assignment allows us to conclude cause-and-effect" is true of both designs. Similarly, "this allows the dermatologist to make conclusions about people with differing acne severity" is true of both designs. Also, "reduces bias" and "reduces variability in the estimates of the individual treatment means" is true of neither design.
- Responses that describe only the set-up of a matched-pairs experiment do not satisfy the requirement to describe an advantage of a matched-pairs design. For example, the response "in a matched-pairs design, the members of each pair will be similar in terms of acne severity" does not describe an advantage. However, "in a matched-pairs design, we can compare two people with similar acne severity" does describe an advantage.
- Advantages of a matched-pairs design that satisfy component 1 include "makes it easier to determine if the drug is effective," "gives a better estimate of the effect of the new drug," "reduces variability in the estimate of the drug effect," "makes the difference between the drug and the placebo more easily distinguishable," and "gives a clearer picture of how well the drug works."
- Advantages of a matched-pairs design that don't satisfy component 1 include "accounts for a source of variability," "controls for potentially confounding variables," "allows you to distinguish variation due to severity from variation due to treatment," "each person can be compared to someone similar," "reduces variability," "more balanced treatment groups," and "more accurate results."

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- It is acceptable to provide a disadvantage of a completely randomized design rather than an advantage of the matched-pairs design (e.g., "The completely randomized design will make it harder to find convincing evidence that the new drug is better").
- It is acceptable to use the term "blocking" as a synonym for "pairing."
- A response that states that a matched-pairs design requires a smaller sample size to get power or precision equal to that in a completely randomized design and describes this advantage in context should be scored E.

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	Model Solution	Scoring
(c)	For each pair of twins, label one person as twin A and label the other person as twin B. For each pair of twins, toss a coin. If the coin lands on heads, twin A gets the placebo and twin B gets the active drug. If the coin lands on tails, twin A gets the active drug and twin B gets the placebo. *OR* Label the members of each pair of twins as "Twin 1" and "Twin 2." Using a random number generator, generate an integer from 1 to 2. Give the drug to the twin whose number is selected and the placebo to the twin whose number is not selected. Repeat for all pairs of twins. *OR* Label 1 notecard "A" and another notecard "B." For each pair of twins, shuffle the cards and give one card to each twin. The twin who gets "A" receives the drug and the twin who gets "B" receives the placebo.	**Essentially correct (E)** if the response randomly assigns the two treatments within pairs of twins AND satisfies the following three components: 1. Uses a random process (e.g., flipping a coin, using a random number generator, shuffling cards) that gives each twin in a pair a 50% probability of getting the drug and a 50% probability of getting the placebo 2. Describes how to use the random process to assign one specific twin in each pair to the drug and the other twin to the placebo 3. Indicates that the random assignment process will be completed for each pair of twins **Partially correct (P)** if the response randomly assigns the two treatments within pairs of twins AND satisfies only two of the three components for E. **Incorrect (I)** if the response does not satisfy the criteria for E or P.

Additional Notes:

- A response that does not randomly assign both treatments within pairs of twins should be scored incorrect (I). Examples include a response that describes a completely randomized design, describes a crossover design where each person receives both treatments, uses pairs other than twins, does not use random assignment, or indicates that both twins in a pair receive the same treatment.
- For responses that use slips of paper or selecting items from a hat, the slips must be shuffled (or blindly drawn) or the hat mixed or shaken to have a random process and satisfy component 1.
- To satisfy component 2, the response must describe what to do for each possible outcome of the random process and specify which treatment each twin receives. For example, none of the following descriptions satisfy component 2:
  - "Roll a die. If it is 1–3, give the first twin the drug and the second twin the placebo." (Response doesn't describe what to do if the die is 4–6.)
  - "Have one member of each pair flip a coin. If it is heads, that twin gets the drug. If it is tails, that twin gets the placebo." (Response doesn't indicate what treatment the other twin will receive.)
  - "Flip a coin. If it is heads, give one twin the drug and the other twin the placebo. If it is tails, do the reverse." (Response doesn't specify which twin is getting the drug.)
  - "Label one slip of paper "A" and a second slip of paper "B." Mix them in a hat and have each member of the pair choose one slip." (Response doesn't specify if A represents the new drug or the placebo.)

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- Ignore any discussion about randomly selecting 36 pairs of twins to obtain subjects for the experiment. Likewise, ignore any discussion about how to perform the analysis for a paired design (e.g., "subtract the improvement scores for each pair of twins").
- It is acceptable to refer to each pair of twins as a block.